

ENERGY FROM THE OCEANS

The ocean is always on the move. Storm waves can unleash enough energy to smash concrete, and tides and currents shift vast quantities of water around the globe. The oceans are swept by some of the most powerful winds on Earth. All these forces could be used to generate electricity in ways that do not pollute the environment. For decades we have relied on energy produced by burning coal, oil, and gas, which release gases that cause climate change. As these old technologies are phased out, the oceans will become a major source of energy.

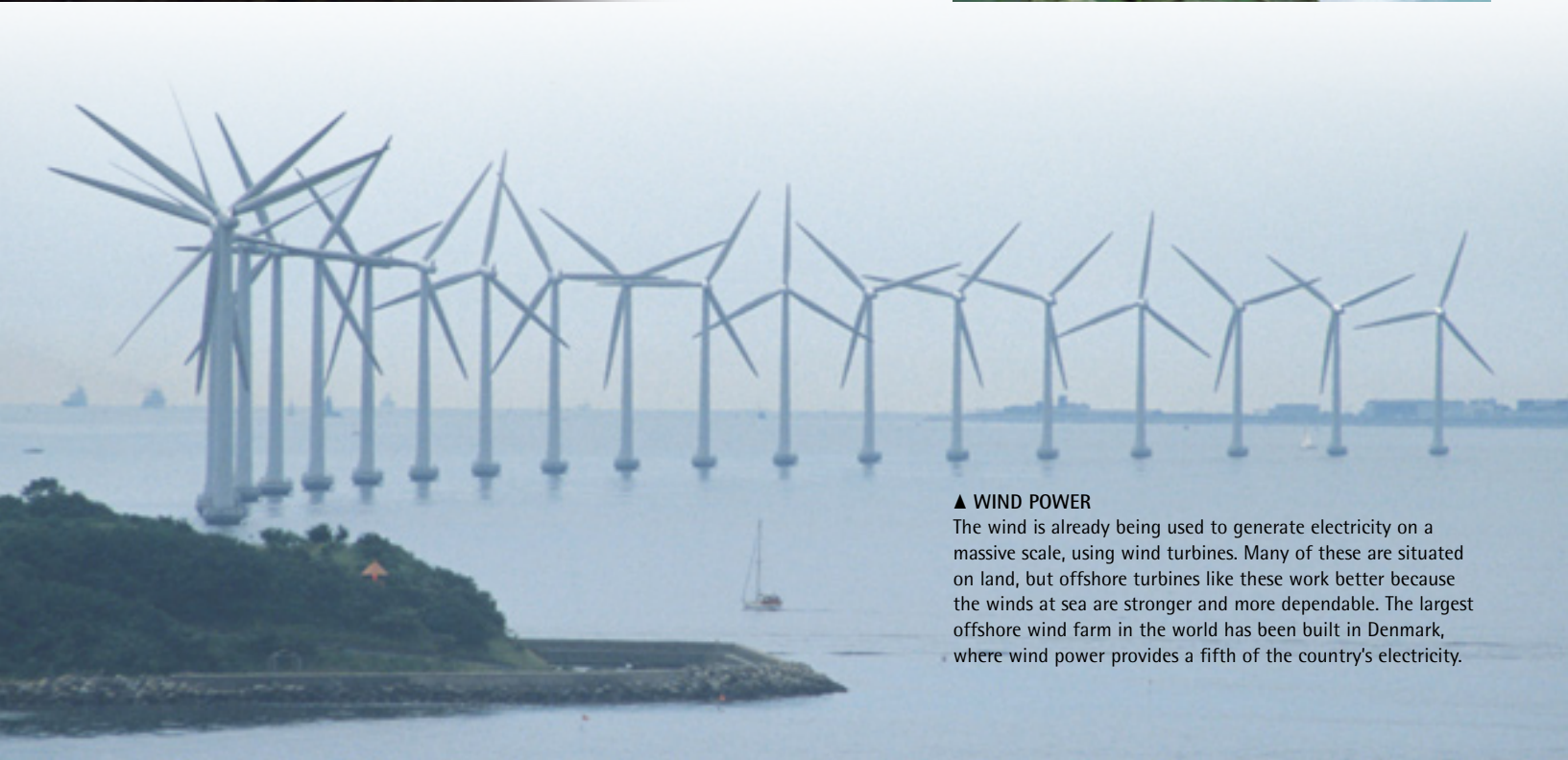
◀ LOST ARTS

Before coal power was widely adopted in the late 18th century, all energy was renewable. Even iron was produced using water power and charcoal made from renewable lumber. Ships were driven by the wind, and tide-powered mills were common. We need to reinvent these technologies for a new age.



▶ WAVE ENERGY

Waves are an obvious source of energy, but it is difficult to turn them into reliable power. This plant in Scotland converts the force of the waves into air pressure, which drives two turbines connected to electricity generators. The system works well, but only because the site is exposed to big waves throughout the year.



▲ WIND POWER

The wind is already being used to generate electricity on a massive scale, using wind turbines. Many of these are situated on land, but offshore turbines like these work better because the winds at sea are stronger and more dependable. The largest offshore wind farm in the world has been built in Denmark, where wind power provides a fifth of the country's electricity.